Shuaipeng (Frank) Dong

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RESEARCH INTERESTS

My research interests include computer vision, continual learning, and knowledge distillation, with a focus on class-incremental visual recognition and memory-efficient lifelong learning for image classification and object detection.

EDUCATION

The Pennsylvania State University University Park, PA
B.S. in Computer Science · B.S. in Mathematics (Systems Analysis) *May 2023*
Cumulative GPA: 3.55 / 4.00 · Dean's List (multiple semesters) · College Honors Student

PUBLICATIONS & PATENTS

Conference Publications

[1] **S. Dong** and D. Whitefield, "Video Bandwidth Prediction by GEO Satellite Terminals for QoE," *41st International Communications Satellite Systems Conference (ICSSC)*, vol. 2024, pp. 26–31, 2024.

[2] **S. Dong**, "Predicting Invasive Ductal Carcinoma Using a Deep Convolutional Neural Network," in *Proc. 3rd International Conference on Artificial Intelligence and Education (ICAIE)*, pp. 191–196, 2022.

Patents

[3] **S. Dong**, "Bandwidth Prediction Using Machine Learning," U.S. Patent Application No. 19/081,355, filed Mar. 2025 (*pending*).

RESEARCH EXPERIENCE

Knowledge Distillation for Class-Incremental Visual Recognition 2026 – Present
Independent Research · Computer Vision / Continual Learning

- Studying catastrophic forgetting in class-incremental object detection on the COCO benchmark, where a single detector must absorb new object categories sequentially while retaining prior ones—framing the problem as a measurable stability–plasticity trade-off.
- Reproducing and dissecting continual-learning and incremental-detection baselines—Learning without Forgetting (LwF), iCaRL, Elastic Response Distillation (ERD), and YOLO-IOD—to isolate where and why prior-class knowledge degrades under sequential class addition.
- Designing teacher–student schemes that distill logit- and feature-level knowledge from a frozen prior-task detector, testing the hypothesis that response and feature distillation preserve old-class accuracy without large exemplar memories.
- Rebuilding the YOLO-World open-vocabulary detection backbone with newer-generation YOLO architectures to test whether stronger feature extractors improve detection performance on COCO under the incremental protocol.
- Redesigning the standard knowledge-distillation loss to remain robust to the teacher detector's false pseudo-labels, easing the gradient explosion that these erroneous old-class targets induce during student training.
- Quantifying retention and plasticity under standard incremental-detection protocols on COCO—per-step and average mAP, forgetting, and backward/forward transfer; manuscript in preparation.

Deep Convolutional Neural Networks for Medical Image Classification 2022
Undergraduate Research · The Pennsylvania State University

- Investigated whether a parameter-efficient convolutional design could detect invasive ductal carcinoma (IDC) from histopathology while remaining robust to the severe class imbalance characteristic of medical-imaging data.
- Designed a depth-wise separable CNN (SeparableConv2D) over 50×50 patches in TensorFlow/Keras—SeparableConv2D–BatchNorm–MaxPooling–Dropout blocks feeding a 256-unit dense softmax classifier trained with Adagrad under binary cross-entropy—paired with a reproducible re-balancing and preprocessing-to-evaluation pipeline.
- Demonstrated that the separable-convolution architecture matched standard convolutional baselines at substantially lower parameter count and computational cost, evidencing a favorable accuracy–efficiency trade-off on imbalanced data.
- First-author publication at ICAIE 2022.

PROFESSIONAL & APPLIED RESEARCH EXPERIENCE

Hughes Network Systems June 2023 – Dec. 2025

Member of Technical Staff II, Software Engineering · Germantown, MD

- Framed satellite rain-fade prediction as a spatiotemporal learning problem and designed a deep 3D-CNN over highly imbalanced radar-imaging and telemetry time-series, achieving >90% classification accuracy with high precision and recall and improving downstream link reliability.
- Hypothesized that targeted synthetic augmentation would counter severe class imbalance; constructed customized synthetic time-series with tailored augmentation that improved F1 and recall by more than 5% and reduced performance variance across weather regimes.
- Developed an LSTM bandwidth-prediction model with a custom loss over streaming TCP-packet data, yielding a peer-reviewed publication (ICSSC 2024) and a pending U.S. patent.
- Built end-to-end ML pipelines (ingestion, automated quality validation, feature generation, real-time inference) and ran distributed multi-GPU / multi-node training (Docker, Kubernetes) to support large-scale experimentation.

NextTier June 2024 – Sept. 2024

Software Engineer Intern — AI Reasoning Agents

- Investigated agentic LLM architectures for multi-step query resolution, building a multi-agent system with autonomous reasoning, dynamic tool use (retrieval, web search, code execution), and task decomposition, grounded by a retrieval-augmented generation (RAG) pipeline over domain knowledge.
- Designed an evaluation framework measuring task success rate, tool-call efficiency, failure-recovery rate, and multi-step completion accuracy, enabling systematic comparison of agent configurations and quantifying gains in factual grounding.

SELECTED PROJECTS

Residual Network — ResNet-50 Implemented from Scratch View Code ↗

Personal Project · TensorFlow / Keras

- Reconstructed the ResNet-50 architecture (~23.6M parameters) from first principles—custom identity and projection residual blocks with skip connections and batch normalization—to probe how residual connections shape optimization and gradient flow in very deep networks.
- Trained and evaluated on a six-class hand-sign image-classification task, reaching ~87.5% test accuracy and empirically confirming the optimization advantage of residual learning over plain deep stacks.

Stock-Price Forecasting with Stacked LSTMs View Code ↗

Personal Project · TensorFlow / Keras

- Studied sequence modeling for noisy financial time series, building a stacked four-layer LSTM (50→60→80→120 units) with dropout regularization to forecast next-step prices for Google (GOOG) and Oracle (ORCL) from multivariate OHLCV inputs.
- Engineered the sequence-windowing and normalization pipeline and trained per-ticker models under a mean-squared-error objective, comparing model variants to assess generalization across the two equities.

RELEVANT COURSEWORK

Machine Learning & AI: Machine Learning & Artificial Intelligence (A−)

Mathematics & Statistics: Linear Algebra (A), Real Analysis (A), Probability Theory (A), Linear Programming (A−), Matrices (A), Multivariable & Vector Calculus (A), Differential Equations (A−), Mathematical Statistics, Discrete Mathematics, Graph Theory

Computer Science: Programming Language Concepts (A−), Digital Logic Design (A−), Numerical Analysis I & II, Data Structures & Algorithms, Theory of Computation, Operating Systems, Systems Programming

TECHNICAL SKILLS

Languages: Python, C++, C, Java, MATLAB

ML / DL: PyTorch, TensorFlow, Keras, MMDetection, OpenCV, Scikit-learn, NumPy, Pandas; CNNs, 3D-CNNs, LSTMs; object detection (YOLO, YOLO-World), knowledge distillation, continual learning

Systems / Tools: Linux, Docker, Kubernetes, Git, FastAPI, PostgreSQL, Jenkins